Pancake Patrollers

Phase 1 AS-IS/TO-BE Scenarios

AS-IS Scenario

- Stevie is trying to go to his next classroom. He knows he needs to walk ahead a few steps, and then turn left around the corner
- However, he is not sure when to turn.
- He took a guess, but turned too early, hit the wall and hurt his head

TO-BE Scenario

- The THEIA app asks Stevie to give his current location and the destinations
- The app calculates the route from the current location to the destination.
- THEIA tells Stevie to “walk ahead 10 steps, then turn left””

AS-IS Scenario

- Calvin is a blind person walking through a hallway.
- Someone has left a chair in his path.
- Calvin may not notice the obstacle until he is close to it, which increases his chances of tripping on it and injuring himself.

TO-BE Scenario

- While Calvin is walking through the hallway with the THEIA app, the app uses available sensors to detect the obstacle.
- The app will then give an audio warning to Calvin
- The app will then recalculate and provide instructions to help Calvin avoid the obstacle.

AS-IS Scenario

- Mable is a blind person who needs to reach another floor of a building.
- She may struggle to locate the elevator or stairs and determine which floor she is on.

TO-BE Scenario

- Mable tells THEIA which room she wants to reach.
- The app determines that the destination is on another floor and provides directions to an elevator or the stairs.
- After reaching the correct floor, THEIA continues giving voice directions to the destination.

AS-IS Scenario

- Emily is a blind person who is lost in an unfamiliar building.
- She may have difficulty explaining her location to someone.

TO-BE Scenario

- THEIA detects that Emily is lost.
- The app asks whether she is in need of assistance.
- If she asks for help or fails to respond, THEIA contacts a designated emergency contact and shares Emilys location with them.

Top Priority Scenario: Indoor Navigation

- What if THEIA cannot determine the users location?
  - Ask the user to move to an identifiable location
  - Attempt the determine the location again
  - Notify the user that navigation is temporarily unavailable
- What if the user takes a wrong turn?
  - Detect that the user has moved away from the planned route
  - Recalculate a route from the users new location
  - Provide updated voice directions

- Who initiates the navigation process?
  - The user manually enters a destination
  - The user gives a voice command
  - THEIA suggests a destination based on the user’s schedule and the user confirms it
- Who can configure the user’s navigation preferences
  - The user
  - A caretaker
  - An authorized accessibility support staff member

- What kinds of destinations should THEIA support
  - Classrooms and lecture halls
  - Offices and labs
  - Restrooms and elevators
  - Other locations within a supported building
- What kinds of routes should THEIA provide
  - Fastest available route
  - Accessible route that avoids stairs

- When should THEIA provide navigation instructions?
  - When navigation begins
  - Before the user needs to turn
  - When the user approaches the destination
  - When the user moves away from the planned route
- When should THEIA recalculate the route?
  - When the user takes a wrong turn
  - When the planned route becomes blocked
  - When the user requests a different route

- How should the user enter their destination?
  - Voice commands
  - Text input
  - Selecting previously saved destinations
- How should THEIA communicate directions?
  - Spoken step by step instructions
  - Audio alerts for important navigation events
  - Vibration feedback for warnings
- How should THEIA handle navigation failures?
  - Attempt to determine the user’s location again
  - Ask the user to confirm their surroundings (if possible)
  - Offer an option to contact a designated emergency contact